

Sensors that smell trouble

RoboticWares developed olfactory sensors to keep track of the amount of LPG in the air. The sensor is an electronic circuit that carries chemicals that react with LPG in the air. Based on this, the electrochemical sensor gives an analogue value output as per the density of the LPG.

Once the amount of LPG reaches 1000 parts per million (ppm), the system turns the alarm 'on' and sends alert SMS to the registered mobile numbers. The alarm switches off only when the amount of LPG in the air reduces to a safe level. SMS is sent once again to the mobile numbers after the leakage stops.

However, they soon realised a buzzer alarm was of no use in an unoccupied house. The alarm was improvised into an SMS-based feedback system.

The first version of Suraksha was developed with an LCD screen and a keypad. For feeding the mobile numbers, the users had to follow certain instructions. In order to simplify the user experience, the team decided to let go of the LCD screen and keypad while creating the SMS feature in the final version. This version lets users register their mobile numbers using the phone's keypad and screen.

The team also found it challenging to design a power circuit

that works properly with all the peripherals. Kumar explains, "When the sensor detects the LPG leak, it sinks in a huge amount of current. At the same time, you have to keep the GSM modem active to send an SMS and buzzer for the audio alarm. Sometimes, the GSM module will not respond properly, and it was a great challenge to work with the power circuits and the noise they generated during active condition."

When the team looked into it, they found that the voltage regulator was not suitable for this device due to its high current consumption. So they modified it to a new regulator

that consumes only 3 amperes.

Making the device affordable for all

Suraksha is currently priced at Rs 4200 with no extra cost for installation. The device has already undergone successful pilot testing at several three-star hotels in Bhubaneswar, Orissa. Now RoboticWares is working on ways to make the device affordable for all Indian homes.

"Since the accidents caused by LPG leakage occur everywhere, we want to make this device cheaper and accessible to rural people," Kumar reveals.

However, RoboticWares has not been able to find investors or government patronage to realise their dream of commercially producing Suraksha. "We are waiting for a government subsidy to reduce the price of the device and introduce it to all Indian homes," Kumar concludes. ●

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